Regression Analysis of County-Level Life Expectancy in the U.S

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Abstract

This report studies the relationship between county-level life expectancy in the United States and 15 socioeconomic, behavioral, environmental, and health-related predictors using the County Health Rankings dataset. After an exploratory data analysis, we fitted a multiple linear regression model without the state variable. Four non-significant predictors (unemployment, PM2.5, primary care physicians, homeownership) and three highly influential outliers were removed, and a further non-significant predictor (gender pay gap) was dropped, yielding a final model with 10 significant predictors that explains 79.2% of the variation in life expectancy. As the Breusch-Pagan test indicated heteroscedasticity, inference was based on HC2 robust standard errors. We compared this model with a lasso-selected model of the same size that differs only in replacing physical inactivity with gender pay gap; the final model performs slightly better on every criterion: cross-validation RMSE, adjusted R^2, AIC, BIC, and test RMSE. Diabetes prevalence, median income, severe housing cost burden, food insecurity, and income inequality emerge as the most important predictors. Limitations include the exclusion of state effects, mild collinearity and the lack of flexible nonlinear modeling.

Exploratory Data Analysis

Set seed, import packages and data set:

```
group_seed <- 29
library(tidyverse)
library(lmtest)
library(caret)
library(glmnet)
library(leaps)
library(MASS)
library(faraway)
library(patchwork)
dat <- read.csv("County_Health_Rankings_county_level_data.csv")
dat <- dat[-1, ]
```

Split the data and check sizes:

```
## [1] "Training set sample size: 928"

## [1] "Test set sample size: 232"

## [1] "Number of predictors: 16"
```

The Only categorical variable is 'state', which includes all states in the US and Washington DC.

For numerical variables, we need to find out their correlation with Life Expectancy:

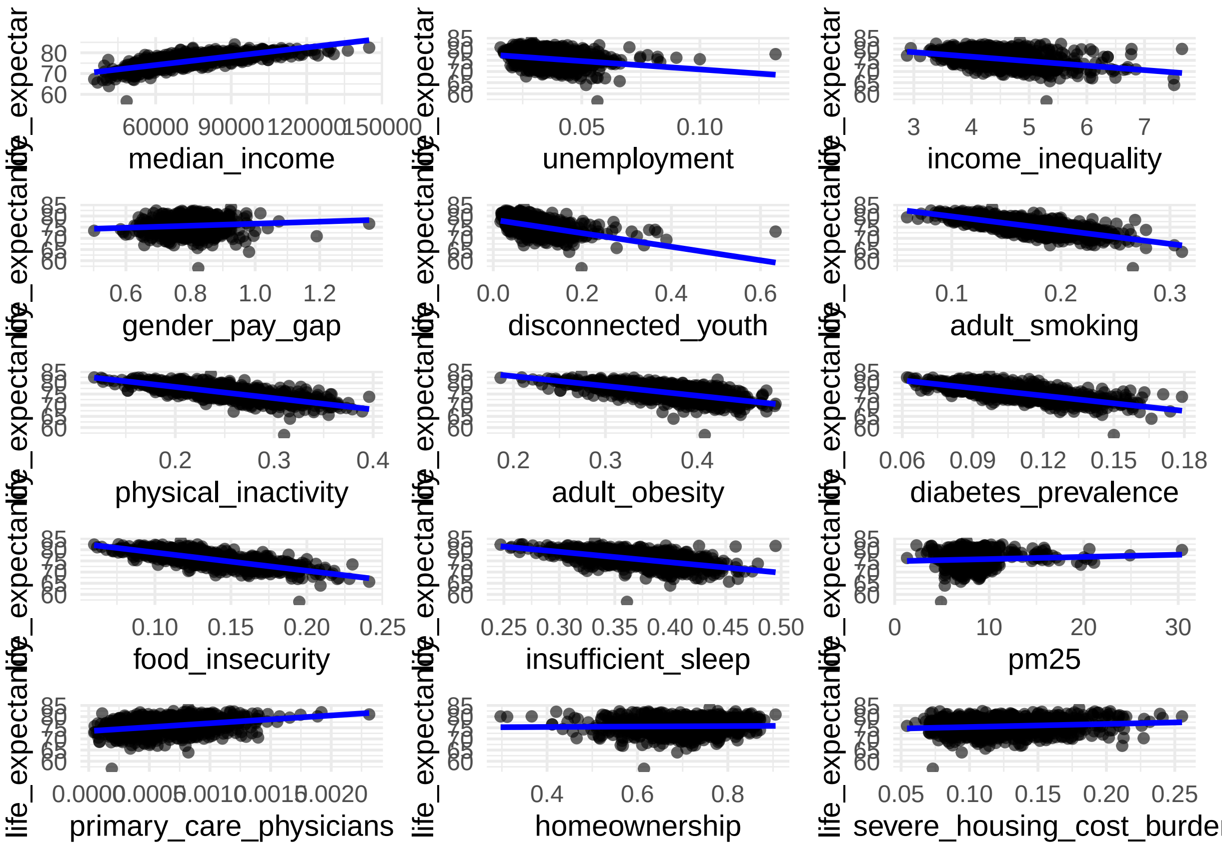


Table 1: Marginal correlations with life expectancy

variable	correlation
food_insecurity	-0.773
median_income	0.772
physical_inactivity	-0.756
adult_smoking	-0.744
diabetes_prevalence	-0.720
adult_obesity	-0.658
insufficient_sleep	-0.570
disconnected_youth	-0.495
income_inequality	-0.406
primary_care_physicians	0.341
unemployment	-0.247
severe_housing_cost_burden	0.142
gender_pay_gap	0.100
pm25	0.072
homeownership	0.026

We find that Median_income, Adult_smoking, physical_inactivity, , diabetes_prevalence, food_insecurity are highly correlated to life_expectancy ($|r| > 0.7$). Specifically, only Median_income is positive correlated to Life_expectancy and others are negatively correlated.

Next we need to find the correlation between the predictors.

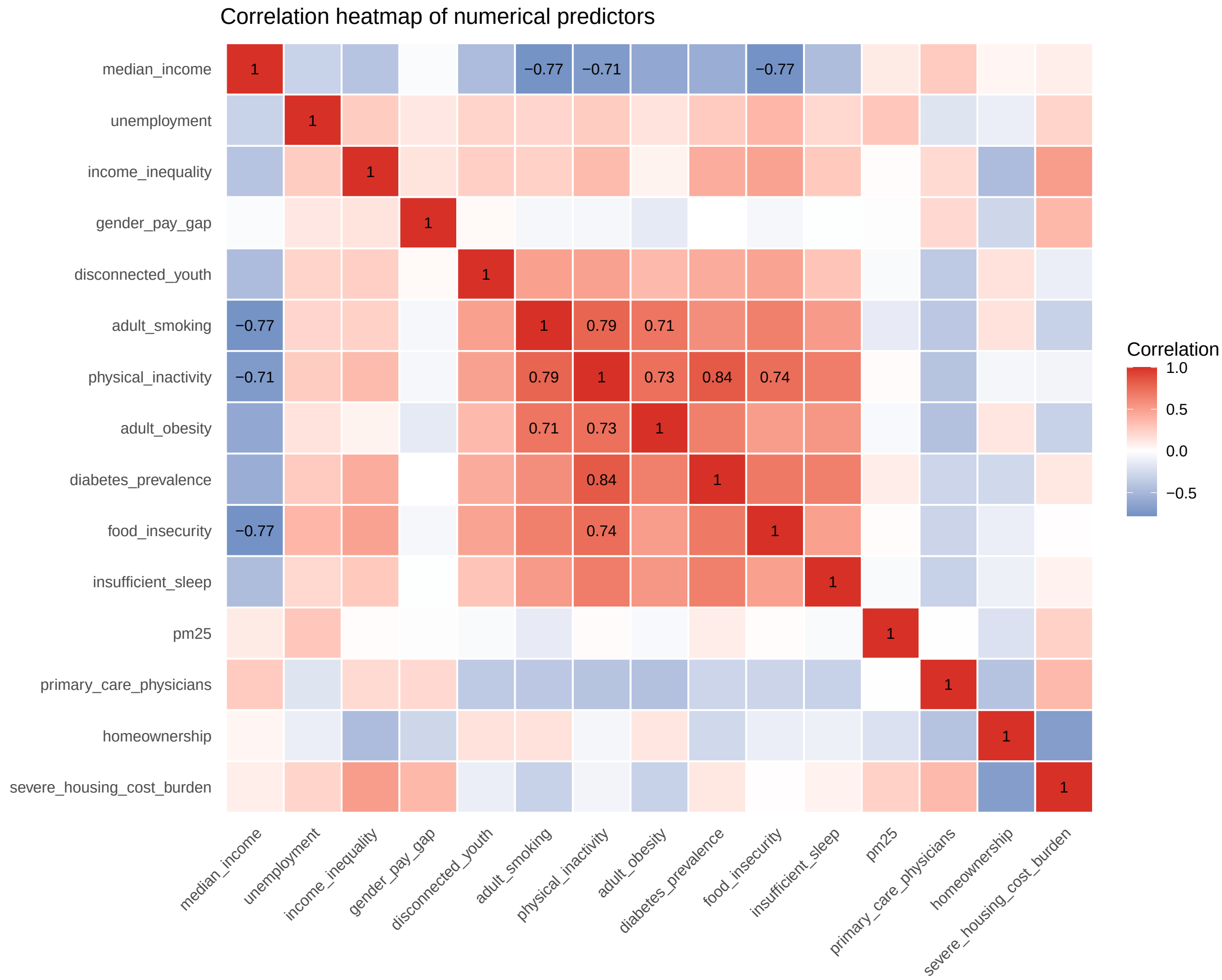
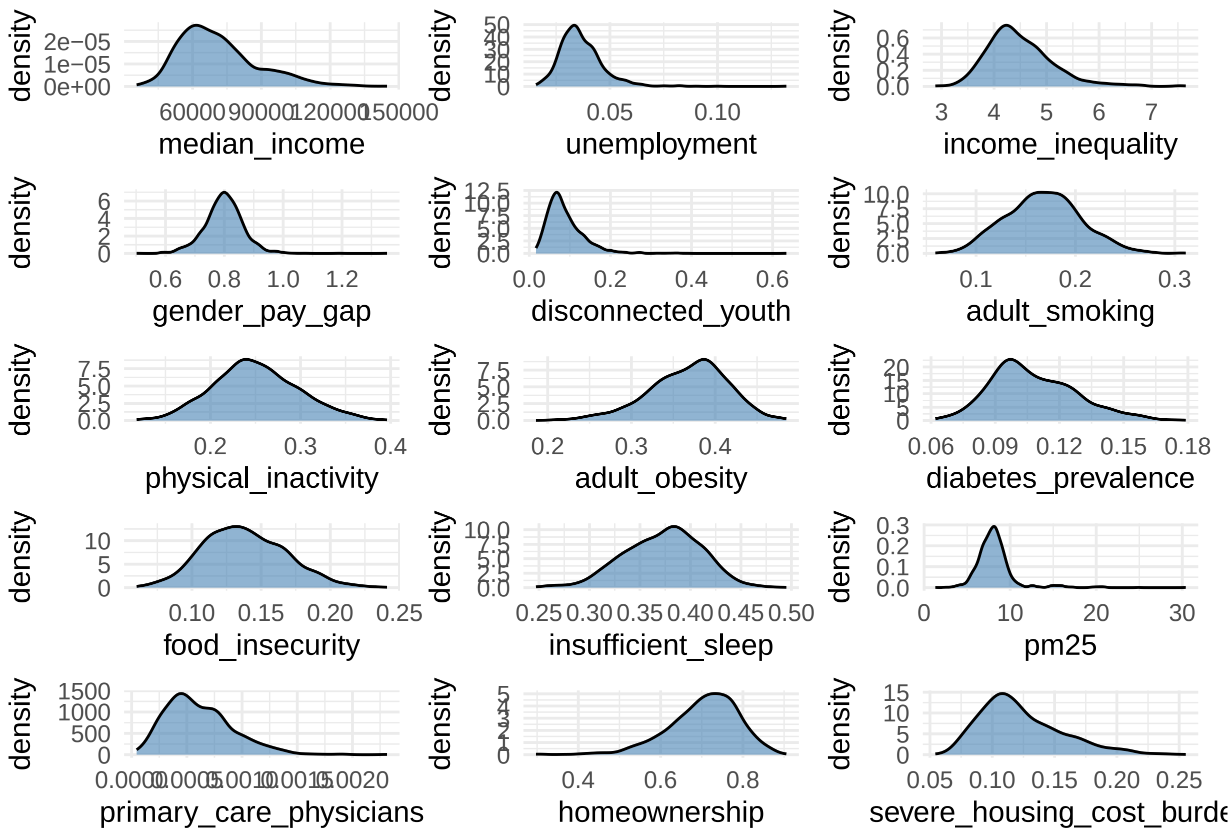


Table 2: Some of highly correlated predictors are shown on this table with |r| > 0.7

Variable1	Variable2	Correlation
adult_smoking	median_income	-0.771
physical_inactivity	median_income	-0.713
food_insecurity	median_income	-0.775
physical_inactivity	adult_smoking	0.788
adult_obesity	adult_smoking	0.708
adult_obesity	physical_inactivity	0.732
diabetes_prevalence	physical_inactivity	0.837
food_insecurity	physical_inactivity	0.743

We find eight pairs of predictors like physical_inactivity and food_insecurity, adult_smoking and median_income are highly correlated with each other.



No nonlinearity can be seen among these predictors based on the scatter plot. Unemployment, disconnected_youth, pm25, median_income and primary_care_physicians are seemed right skewed. homeownership, adult_obesity, insufficient_sleep are left skewed. Outliers can be seen from the scatter plot.

Decision Note:

by EDA we can say that, first, five predictors, physical_inactivity, adult_smoking, food_insecurity, diabetes_prevalence and median_income, have the strong marginal associations with life expectancy. Second, eight pairs of predictors have correlations with each other. Third, we can say that there are skewness of some predictors and outliers, but we can't confirm nonlinearity only based on scatter plot.

Model Building and Diagnostics

We chose to drop state variable and do MLR,

```r
train_dat$state <- NULL
test_dat$state <- NULL # here we just remove state column from both train and test data
```

```r
mod0 <- lm(life_expectancy ~., data = train_dat) # Fit MLR with all numeric variables
summary(mod0)
```

```
##
## Call:
```

```
## lm(formula = life_expectancy ~ ., data = train_dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.4429  -0.8817  -0.0385   0.8901   6.9263
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                8.317e+01  1.683e+00  49.427  < 2e-16 ***
## median_income              4.627e-05  5.888e-06   7.859 1.09e-14 ***
## unemployment               8.350e+00  5.656e+00   1.476 0.140257
## income_inequality         -6.376e-01  1.200e-01  -5.315 1.34e-07 ***
## gender_pay_gap             2.101e+00  8.035e-01   2.615 0.009065 **
## disconnected_youth        -2.520e+00  1.222e+00  -2.063 0.039412 *
## adult_smoking             -1.227e+01  2.829e+00  -4.336 1.61e-05 ***
## physical_inactivity        9.357e+00  2.771e+00   3.377 0.000765 ***
## adult_obesity             -4.292e+00  1.948e+00  -2.204 0.027806 *
## diabetes_prevalence       -4.570e+01  5.791e+00  -7.891 8.59e-15 ***
## food_insecurity           -2.391e+01  3.299e+00  -7.248 9.01e-13 ***
## insufficient_sleep        -7.219e+00  1.862e+00  -3.877 0.000113 ***
## pm25                       1.340e-02  2.373e-02   0.565 0.572500
## primary_care_physicians    2.855e+02  2.187e+02   1.305 0.192112
## homeownership              7.396e-01  9.266e-01   0.798 0.424959
## severe_housing_cost_burden  1.394e+01  2.731e+00   5.106 4.01e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.489 on 912 degrees of freedom
## Multiple R-squared:  0.7768, Adjusted R-squared:  0.7731
## F-statistic: 211.6 on 15 and 912 DF,  p-value: < 2.2e-16
```

So the summary tells us that this model is overall significant with p-value < 2.2e-16. 15 numerical predictors explained about 77.3% of Life expectancy. Also, it tells us that unemployment, pm25, primary_care_physicians and homeownership are not significant. So here we drop all insignificant predictors.

```
mod1 <- lm(life_expectancy ~ . - unemployment  - pm25 - homeownership - primary_care_physicians,
           data = train_dat)

anova(mod0, mod1)$'Pr(>F)'[2]
```

```
## [1] 0.3274602
```

The reduced model obtained by removing the insignificant predictors (unemployment, pm25, primary_care_physicians and homeownership) was compared to the original model. The ANOVA results show that the reduction does not significantly worsen model fit (p = 0.3275). Therefore, the simpler model is preferred for its improved interpretability.

Here we find high leverage points.

```
cutoff_leverage <- 2 * p / n # Rule of thumb: 2p/n
high_lev <- which(lev > cutoff_leverage)
```

```
## There are 57 high leverage points
```

```r
cat("Potential outliers:\n")
print(data.frame(stud_res = stud_res[outliers]))
```

```
## Potential outliers:
##      stud_res
## 203  4.339289
## 508  4.866348
## 601 -8.892197
```

So we find three outliers: 203, 508, 601.

Here we are testing whether outliers are also high leverage points.

```r
overlap <- intersect(high_lev, outliers)
overlap
```

```
## [1] 203 508 601
```

This confirms outliers we found are also high leverage points.

Here we are going to find high influential points using cook's distance

```r
overlap <- intersect(overlap, which(cook > 4/n))
```

```
## [1] 203 508 601
```

So cook's distance confirms that 203, 508, and 601 are high influential points (Cook's distance > 4/n).

Remove Outliers and fit a Model

```r
remove_ids <- c(203, 508, 601)
train_clean <- train_dat[-remove_ids, ]
mod2 <- lm(life_expectancy ~ . - unemployment  - pm25 - homeownership - primary_care_physicians,
           data = train_clean)

summary(mod2)
```

```
##
## Call:
## lm(formula = life_expectancy ~ . - unemployment - pm25 - homeownership -
##     primary_care_physicians, data = train_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.6079 -0.8980 -0.0523  0.9030  4.6328
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   8.560e+01  1.347e+00  63.563  < 2e-16 ***
```

```
## median_income              4.649e-05  5.462e-06   8.511  < 2e-16 ***
## income_inequality         -6.797e-01  1.083e-01  -6.276 5.35e-10 ***
## gender_pay_gap             1.209e+00  7.757e-01   1.558 0.119486    
## disconnected_youth        -3.110e+00  1.093e+00  -2.845 0.004547 ** 
## adult_smoking             -1.018e+01  2.645e+00  -3.850 0.000126 ***
## physical_inactivity        7.550e+00  2.560e+00   2.949 0.003266 ** 
## adult_obesity             -5.209e+00  1.829e+00  -2.848 0.004496 ** 
## diabetes_prevalence       -4.388e+01  5.277e+00  -8.316 3.28e-16 ***
## food_insecurity           -2.165e+01  3.030e+00  -7.146 1.82e-12 ***
## insufficient_sleep        -8.695e+00  1.716e+00  -5.067 4.88e-07 ***
## severe_housing_cost_burden  1.477e+01  2.235e+00   6.609 6.57e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.397 on 913 degrees of freedom
## Multiple R-squared:  0.7949, Adjusted R-squared:  0.7925 
## F-statistic: 321.8 on 11 and 913 DF,  p-value: < 2.2e-16
```

So the summary tells us that after removing outliers the model is overall significant with p-value < 2.2e-16. 11 numerical predictors explained about 79.3% of Life expectancy. Also, it tells us that gender_pay_gap is no longer significant after removing outliers.

```r
mod3 <- lm(life_expectancy ~ . - unemployment  - pm25 - homeownership
           - primary_care_physicians -gender_pay_gap, data = train_clean)
cat("So it's safe to drop gender_pay_gap since p value is", anova(mod2, mod3)$'Pr(>F)'[2])
```

```
## So it's safe to drop gender_pay_gap since p value is 0.1194862
```

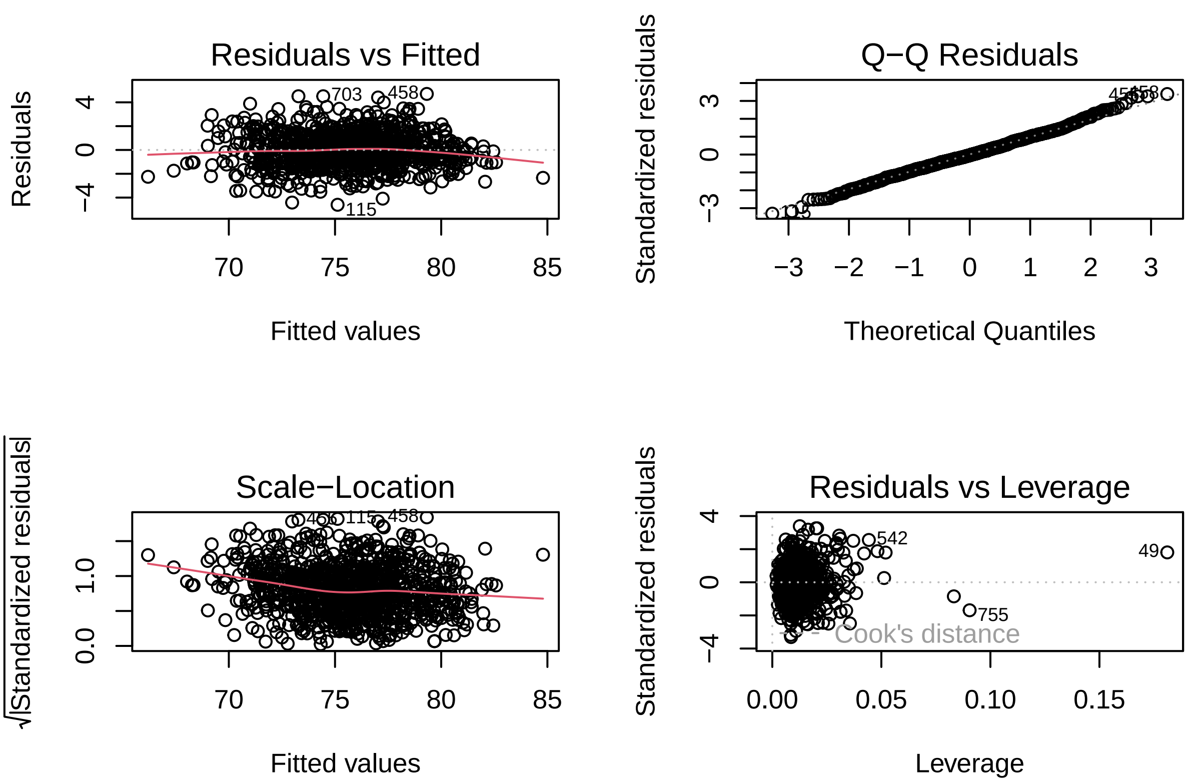
```r
summary(mod3)
```

```
##
## Call:
## lm(formula = life_expectancy ~ . - unemployment - pm25 - homeownership -
##     primary_care_physicians - gender_pay_gap, data = train_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -4.6027 -0.8922 -0.0547  0.9281  4.7052 
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)    
## (Intercept)                8.676e+01  1.123e+00  77.295  < 2e-16 ***
## median_income              4.521e-05  5.405e-06   8.365 2.23e-16 ***
## income_inequality         -7.018e-01  1.075e-01  -6.530 1.09e-10 ***
## disconnected_youth        -2.978e+00  1.091e+00  -2.730 0.006459 ** 
## adult_smoking             -9.733e+00  2.631e+00  -3.699 0.000229 ***
## physical_inactivity        7.387e+00  2.560e+00   2.886 0.003992 ** 
## adult_obesity             -5.521e+00  1.819e+00  -3.035 0.002473 ** 
## diabetes_prevalence       -4.363e+01  5.278e+00  -8.266 4.83e-16 ***
## food_insecurity           -2.233e+01  3.001e+00  -7.440 2.32e-13 ***
## insufficient_sleep        -8.754e+00  1.717e+00  -5.099 4.15e-07 ***
## severe_housing_cost_burden  1.599e+01  2.095e+00   7.630 5.91e-14 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.398 on 914 degrees of freedom
## Multiple R-squared:  0.7944, Adjusted R-squared:  0.7921
## F-statistic: 353.1 on 10 and 914 DF,  p-value: < 2.2e-16
```

So the summary tells us that after removing gender_pay_gap the model is overall significant with p-value < 2.2e-16. 10 numerical predictors explained about 79.2% of Life expectancy.

Diagnostic plots and tests of mod3




```
## Raintest p-value (H0: linear): 0.7214794
## Shapiro-Wilk normality test p-value (H0: normal): 0.07603633
## Count of Cook's distance > 1: 0
## Breusch-Pagan heteroscedasticity test p-value (H0: homoscedastic): 1.91129e-07
## Predictors with VIF > 5: physical_inactivity
##           6.697517
```

```
p3   <- length(coef(mod3))
n3   <- nrow(train_clean)
lev3 <- influence(mod3)$hat
stud_res3 <- rstudent(mod3)
cutoff3 <- 2 * p3 / n3
thr3 <- abs(qt(1 - 0.05/(2 * n3), df = n3 - p3 - 1))
high_lev3 <- which(lev3 > cutoff3)
```

```
outliers3 <- which(abs(stud_res3) > thr3)
cat("High leverage points of mod3:", if(length(high_lev3)) paste(high_lev3, collapse = ", ") else "none
cat("Outliers of mod3:", if(length(outliers3)) paste(outliers3, collapse = ", ") else "none", "\n")
```

```
## High leverage points of mod3: 21, 49, 56, 91, 178, 186, 192, 195, 250, 295, 297, 304, 346, 351, 363,
## Outliers of mod3: none
```

Diagnostic Summary

The diagnostic plots suggest that the model assumptions are reasonable. Based on the plots, the linearity is not a obvious problem since the curve is relatively flat. And raintest(p = 0.7214794) proves linearity. The Shapiro-Wilk test shows that p = 0.07603633 so normality is not a concern. No high influential points with Cook's distance > 1. The p value of Breusch-Pagan test is 1.91129e-07 so heteroscedasticity would be an issue. The multicollinearity problem is not really an issue since there is only one predictor with VIF > 5 (6.697517).

Here we do HC standard errors to address heteroscedasticity issue.

```
library(sandwich)
robust_se <- coeftest(mod3, vcov = vcovHC(mod3, type = "HC2"))
knitr::kable(robust_se[,], caption = "HC2 robust coefficient tests")
```

Table 3: HC2 robust coefficient tests

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	86.7648787	1.1501664	75.436806	0.0000000
median_income	0.0000452	0.0000057	7.999500	0.0000000
income_inequality	-0.7017511	0.1195394	-5.870457	0.0000000
disconnected_youth	-2.9776601	1.3746495	-2.166123	0.0305597
adult_smoking	-9.7334184	3.1721271	-3.068420	0.0022154
physical_inactivity	7.3873118	2.8026108	2.635868	0.0085345
adult_obesity	-5.5211519	1.9121013	-2.887479	0.0039750
diabetes_prevalence	-43.6328062	5.7257994	-7.620387	0.0000000
food_insecurity	-22.3270972	3.4345664	-6.500703	0.0000000
insufficient_sleep	-8.7540987	1.9581095	-4.470689	0.0000088
severe_housing_cost_burden	15.9862746	2.2541650	7.091883	0.0000000

Model Comparison

So we fit model3(mod3) by removing state and other predictors, removing outliers, removing gender_pay_gap. Basically what we did was manual variable selection. So for this part we choose to reduce the model by shrinkage, specifically using lasso.

```
X_clean <- model.matrix(life_expectancy ~ . , data = train_clean)[, -1]
y_clean <- train_clean$life_expectancy
set.seed(group_seed)
cv_lasso <- cv.glmnet(X_clean, y_clean, alpha = 1, nfolds = 10)
```

```
## lambda.min: 0.00682883
```

```
## lambda.1se: 0.1112929
```

```
## The number of predictors left of lasso model(lambda.min) is 14
```

```
## The number of predictors left of lasso model(lambda.1se) is 10
```

Since our mod3 has 10 predictors, we here use lambda.1se so that our lasso model also has 10 predictors. Following are predictors that both models have:

```
## [1] "median_income"              "income_inequality"
## [3] "disconnected_youth"         "adult_smoking"
## [5] "adult_obesity"              "diabetes_prevalence"
## [7] "food_insecurity"            "insufficient_sleep"
## [9] "severe_housing_cost_burden"
```

```
## Only in mod3: physical_inactivity
```

```
## Only in lasso: gender_pay_gap
```

Table 4: Model comparison metrics (CV RMSE, adjusted R^2, AIC, BIC)

Model	CV_RMSE	Adj_R2	AIC	BIC
mod3	1.412	0.792	3258	3316
mod_lasso	1.437	0.791	3264	3322

We compared mod3 (backward selection: 10 predictors including physical_inactivity) and mod_lasso (LASSO with 1se: 10 predictors, swapping physical_inactivity for gender_pay_gap) using four criteria: 10-fold CV RMSE, adjusted R^2, AIC, and BIC. As shown in Table 4, mod3 has a slightly lower CV RMSE (1.412 vs. 1.437), slightly higher adjusted R^2 (0.792 vs. 0.791), and lower AIC (3258 vs. 3264) and BIC (3316 vs. 3322). mod3 is favored on every metric. Notably, the only difference between the two models is that lasso replaces physical_inactivity (highly correlated with food_insecurity) with gender_pay_gap, yet performance is nearly identical, which is further evidence that mild collinearity does not affect the analysis. We therefore select mod3 as our final model.

Conclusion and discussion

In conclusion, our **final model** for this dataset is mod3, that is Life expectancy ~ median_income, income_inequality, disconnected_youth, adult_smoking, physical_inactivity, adult_obesity, diabetes_prevalence, food_insecurity, insufficient_sleep, severe_housing_cost_burden.

Table 5: HC2 robust coefficient tests

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	86.7648787	1.1501664	75.436806	0.0000000
median_income	0.0000452	0.0000057	7.999500	0.0000000
income_inequality	-0.7017511	0.1195394	-5.870457	0.0000000
disconnected_youth	-2.9776601	1.3746495	-2.166123	0.0305597
adult_smoking	-9.7334184	3.1721271	-3.068420	0.0022154

	Estimate	Std. Error	t value	Pr(>|t|)
physical_inactivity	7.3873118	2.8026108	2.635868	0.0085345
adult_obesity	-5.5211519	1.9121013	-2.887479	0.0039750
diabetes_prevalence	-43.6328062	5.7257994	-7.620387	0.0000000
food_insecurity	-22.3270972	3.4345664	-6.500703	0.0000000
insufficient_sleep	-8.7540987	1.9581095	-4.470689	0.0000088
severe_housing_cost_burden	15.9862746	2.2541650	7.091883	0.0000000

Which variables appear to be important?

Based on the HC2 robust inference, all 10 predictors in `mod3` are statistically significant at the 0.05 level. Based on t-values, the five most important predictors are: **`diabetes_prevalence`**($t = -7.62$), **`median_income`** ($t = 8.00$), **`severe_housing_cost_burden`** ($t = 7.09$), **`food_insecurity`** ($t = -6.50$), **`income_inequality`** ($t = -5.87$)

Interpretation of important numerical predictors

We interpret the five key coefficients from the final model (HC2 robust standard errors). - **`diabetes_prevalence`** ($\hat{\beta} = -43.633$): A 1 percentage point increase in diabetes prevalence is associated with a decrease of about $43.633 \times 0.01 \approx 0.436$ years in average life expectancy - **`median_income`** ($\hat{\beta} = 0.0000452$): A \$10,000 increase in county median income is associated with an increase of approximately $0.0000452 \times 10000 = 0.452$ years in average life expectancy. - **`severe_housing_cost_burden`** ($\hat{\beta} = 15.986$): A 1 percentage point increase in severe housing cost burden is associated with an increase of about $15.986 \times 0.01 \approx 0.16$ years. This could be bacause counties with high housing costs tend to be urban areas with better healthcare access. - **`food_insecurity`** ($\hat{\beta} = -22.327$): A 1 percentage point increase in food insecurity is associated with a decrease of about $22.327 \times 0.01 \approx 0.223$ years in average life expectancy. - **`income_inequality`** ($\hat{\beta} = -0.702$): A one-unit increase in the income inequality ratio is associated with a decrease of about 0.702 years in average life expectancy.

Test RMSE

Table 6: Test set prediction errors (in years)

Model	Test.RMSE
mod3 (final)	1.8760
mod_lasso	1.9287

The test RMSE for our final model (mod3) is smaller, which favors mod3 and is consistent with the earlier comparison. Thus, the model selected by manual backward elimination performs at least slightly better than the lasso-guided model. So mod3 is our final choice.

Limitations and open questions

Given there is page limit for this project, we have met following issues. First, our model does not include state. Second, we did not transform the response to directly model the variance structure; instead we mitigated heteroscedasticity by using HC2 robust standard errors for inference. Third, we do not directly address the collinearity issue. However, it is mild (only one predictor with VIF > 5), and the lasso comparison model, which handles collinearity by shrinkage, yields nearly identical performance and conclusions, so collinearity does not undermine the robustness of our findings. Forth, although the raintest and scatter plots support linearity, we did not explore flexible nonlinear forms such as splines or interactions. Fifth, we re-checked mod3 for high leverage points and outliers: no outliers and no points with Cook's distance > 1 were found, though 50 high-leverage points were identified; we chose not to remove them as they do not materially affect coefficient inference. Finally, the predictive error of mod3 is about 1.9, suggesting there are still some unexplained variation for life expectancy.

Additional Regression Method

To further validate our findings, we also fitted a random forest, a non-parametric method that automatically captures nonlinearities and interactions, and is robust to collinearity, skewness, and outliers. The 10-fold CV RMSE and test RMSE are compared with the linear models below.

```r
ctrl_rf <- trainControl(method = "cv", number = 10)
cv_rf <- train(life_expectancy ~ ., data = train_clean, method = "rf", trControl = ctrl_rf)
pred_rf <- predict(cv_rf, newdata = test_dat)
rmse_rf <- sqrt(mean((test_dat$life_expectancy - pred_rf)^2))
```

```
## Random forest CV RMSE: 1.344712 1.319778 1.332309
```

```
## Random forest test RMSE: 1.862416
```

The random forest achieves a slightly lower CV RMSE (best mtry: 1.320 vs. 1.412 for mod3) and a slightly lower test RMSE (1.862 vs. about 1.876 for mod3), indicating that the flexible model captures only a small amount of structure beyond linearity. The linear model therefore remains competitive while being far more interpretable.